# Amalka Dewmini

Undergraduate
dissanayakedewmini636@gmail.com | 0770731032 | Janaka Kumara Lane,Amunudowa,Bandarawela
**Linkedin | GitHub | HackerRank**

## Profile Summary

BSc (Hons) Software Engineering undergraduate at Sri Lanka Technology Campus with a strong interest in Quality Assurance, skilled in software testing undamentals including test case design, functional testing, and defect reporting across the Software Testing Life Cycle. Experienced with manual and tool-assisted testing using automation tool, along with requirement analysis to ensure applications meet functional and quality standards. Builds on a solid software engineering foundation (React, Node.js, SQL) to understand systems end-to-end and identify issues effectively.

## Education

**Sri Lanka Technology Campus(SLTC)** Ingiriya Road ,Padukka
BSc(Hons) in Software Engineering March 2023 - Present

**Vishaka Girls College** Bandarawela,Sri Lanka
G.C.E. Advanced Level -Technology Stream with Information & Communication Technology (ICT) 2021

## Skills

| | |
|---|---|
| Programming Languages: | Python, Java, JavaScrip |
| Web Technologies: | HTML, CSS, JavaScript, React.js, Node.js, Express.js, REST API |
| Testing Tools: | Selenium, Postman, Data Cleaning, Data Interpretation |
| Database: | SQL, MySQL, MongoDB, Pandas and NumPy |
| Testing Skills: | Test case design, Functional testing, Software Testing Life Cycle , Requirement Analysis, Defect reporting |
| Tools: | Figma, Selenium, Git and GitHub, Visual Studio Code, Power BI, Excel |
| Data Analysis: | Power BI, Excel, RFM Segmentation, SHAP, KPI & Trend Reporting |

## Projects / Open-Source

**Explainable AI Loan Approval System**
Technologies Used:Machine Learning, Explainable AI, Full-Stack Development
Developing a human-centered XAI framework for transparent, fair AI-driven loan approval, focused on consumer microloans in developing-country contexts. Training tree-based models (XGBoost, Random Forest) on South Asian credit data with Python/Scikit-Learn, using SHAP and DiCE counterfactuals to generate actionable, human-readable explanations, and AIF360 to evaluate and mitigate bias via Demographic Parity and Equal Opportunity. Building a full-stack app (Flutter front-end, Node.js/Express API), containerized with Docker and deployed via Terraform on AWS.

**SmartCare: AI-Based Disease Risk Classification System | Link**
Technologies Used:Python, Scikit-learn, XGBoost, SHAP, Streamlit
Built a machine learning pipeline for hospital disease-risk classification using Logistic Regression, Random Forest, and XGBoost, with SHAP for model explainability. Identified a fairness gap in risk coverage across age groups through subgroup analysis. Deployed a live prediction prototype using Streamlit and contributed to technical report sections covering problem definition, literature review, and explainable AI analysis as part of a four-member team.

**Node Classification on OGBN-Arxiv using Graph Neural Networks | Link**
Technologies Used: Python, PyTorch, PyTorch Geometric, GCN, GAT, Google Colab
Developed and evaluated Graph Convolutional Network (GCN) and Graph Attention Network (GAT) models for node classification on the OGBN-Arxiv citation dataset, with GPU training in Google Colab. Troubleshot framework compatibility issues during implementation and compared the two architectures through hyperparameter tuning and evaluation metrics.

**Utopia Civic Engagement Platform | Link**

Technologies Used: Node.js, Express, MongoDB, Leaflet

system design for a web-based civic engagement platform connecting citizens, government officers, and administrators for reporting and resolving civic and utility issues. Defined the full report lifecycle submission, validation, routing, assignment, status tracking, messaging, and resolution along with role-based access control, verified public alerts, and audit logging. Modeled the system using UML use case, activity, sequence, class, and object diagrams, and designed the high-level architecture spanning a responsive web front-end, Express API layer, MongoDB data store, and external map/weather service integrations.

Built with Node.js, Express, MongoDB, and Leaflet for map-based reporting.

**Healthcare Management System | Link**

Technologies Used: React, Vite, Web Development

Developed a responsive hospital management system for managing patients, appointments, and medical records through a clean, user-friendly interface. Built with React and Vite, focusing on intuitive navigation and streamlined healthcare workflows for staff and administrators.

**E-Commerce Sales & Customer Analytics Dashboard | Link**

Technologies Used: React, Vite, FastAPI, Pandas, RESTful API Design

Built a full-stack analytics platform for e-commerce sales and customer intelligence. Designed a Pandas-based data pipeline that cleans raw transactional data deduplication, cancellation/return detection, and date normalization before computing revenue, order, and product-level metrics. Implemented RFM (Recency, Frequency, Monetary) customer segmentation with quantile-based scoring to classify customers into actionable segments (Champions, At Risk, Loyal Customers, etc.). Exposed the analytics engine through a FastAPI REST API with endpoints for KPIs, revenue trends (daily/weekly/monthly), top products, category breakdowns, and paginated/filterable customer segmentation. Built a React (Vite) frontend that consumes these endpoints to render an interactive dashboard with date-range and category filtering, revenue trend charts, product/category rankings, and a searchable customer segmentation table.

## Certifications

- Introduction to Software Testing - **Simplilearn**
- Free Product Quality Management Course - **Simplilearn**
- Power BI For Biginers - **Simplilearn**
- HackerRank Software Engineer Certification - **HackerRank**
- SQL (Basic) - **HackerRank**
- Cloud Computing For Absolute Beginners - **CoDeKu DevOps Academy**
- Learn Power BI Data Modeling with DAX - **Simplilearn**

## Honors & Awards

- Deputy Head, Membership Development Subcommittee - IEEE Computer Society Student Branch Chapter of SLTC -2024|2025
- Team Head, Codemania v5.0, GitGenius 2024, and Arduino Challenge 2025